

Skills Summary

Adding and Subtracting Fractions with Like Denominators

Use this reference while completing your worksheet or when studying.

Skill 1 — Adding Fractions with Like Denominators

Rule: $\frac{a}{n} + \frac{b}{n} = \frac{a+b}{n}$

Why: the denominator n names the *size* of one piece. Adding does not re-cut the bar, so the pieces keep their size and only the count changes.

- Add the numerators (top numbers).
- Copy the denominator (bottom number) unchanged.
- If the numerator ends up larger than the denominator, the answer is more than one whole.

Example: $\frac{3}{8} + \frac{2}{8}$

Step 1. Both fractions count eighths, so the pieces are already the same size — just count pieces.

Step 2. 3 eighths and 2 more eighths make $3 + 2 = 5$ eighths, and the bar is still cut in 8:

$$\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$$

Try it: $\frac{4}{9} + \frac{3}{9}$

$\frac{6}{9}$ check

Skill 2 — Subtracting, then Simplifying

Rule: $\frac{a}{n} - \frac{b}{n} = \frac{a-b}{n}$

Simplify: divide the numerator and the denominator by the same number until the only number that divides both is 1. Simplifying renames the amount; it never changes it.

Example: $\frac{7}{10} - \frac{3}{10}$

Step 1. Take away means cross out pieces of the same size, so subtract the counts and keep the tenths.

Step 2. $7 - 3 = 4$ tenths, and 4 and 10 both divide by 2: $\frac{7}{10} - \frac{3}{10} = \frac{4}{10} = \frac{2}{5}$

Try it: $\frac{5}{8} - \frac{1}{8}$, in simplest form

$\frac{2}{4}$ check

Skill 3 — Finding a Missing Numerator

Shape of the question: $\frac{a}{n} + \underline{\hspace{1cm}} = \frac{c}{n}$

Since every piece is an n th, ask “how many more n ths?” and **count up** from a to c . The missing count is $c - a$, written over the same denominator. Always check by adding your answer back.

Example: $\frac{1}{7} + \underline{\hspace{1cm}} = \frac{6}{7}$

Step 1. Both known fractions are sevenths, so the missing piece must be sevenths too — only its numerator is unknown.

Step 2. Count up from 1 seventh to 6 sevenths: $6 - 1 = 5$ more sevenths. Check: $\frac{1}{7} + \frac{5}{7} = \frac{6}{7}$,
so the blank is $\frac{5}{7}$

Try it: $\frac{2}{11} + \underline{\hspace{1cm}} = \frac{9}{11}$

$\frac{11}{2}$ check

(!) **Watch out:** Never add or subtract the denominators. $\frac{3}{7} + \frac{2}{7}$ is $\frac{5}{7}$, not $\frac{5}{14}$ — adding cannot make the total *smaller*, and nothing re-cut the bar into fourteenths.